You

I want to make a random image generator in Javascript as a html webpage. The randomised images should be all be from a subfolder called "pics", there should be a next button that pulls up the next random image and a back button that goes to the previous image and a randomise button that changes the random order. How can I make this?

Copilot

Great, let's get started with creating your random image generator! Here's a basic outline of what we need to do:

1. Create an HTML structure with buttons and an image placeholder.
2. Write JavaScript code to handle the image randomization and navigation.
3. Store your images in a subfolder called "pics".

Here's a sample implementation to get you going:

HTML Structure

Create an HTML file (e.g., index.html) with the following content:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Random Image Generator</title>
    <style>
        #image-container img {
            width: 500px;
            height: auto;
        }
    </style>
</head>
<body>
    <div id="image-container">
        <img id="random-image" src="pics/image1.jpg" alt="Random Image">
    </div>
    <button onclick="showPreviousImage()">Back</button>
    <button onclick="showRandomImage()">Next</button>
    <button onclick="randomizeOrder()">Randomize</button>

    <script src="script.js"></script>
</body>
</html>
```

JavaScript Logic

Create a JavaScript file (e.g., script.js) with the following content:

```
// List of images in the "pics" folder
const images = ["pics/image1.jpg", "pics/image2.jpg", "pics/image3.jpg", "pics/image4.jpg"];
let currentIndex = 0;
let shuffledIndexes = shuffleArray([...Array(images.length).keys()]);

function showImage(index) {
    const imgElement = document.getElementById("random-image");
    imgElement.src = images[index];
}
```

```
function showPreviousImage() {
    currentIndex = (currentIndex - 1 + images.length) % images.length;
    showImage(shuffledIndexes[currentIndex]);
}

function showRandomImage() {
    currentIndex = (currentIndex + 1) % images.length;
    showImage(shuffledIndexes[currentIndex]);
}

function randomizeOrder() {
    shuffledIndexes = shuffleArray([...Array(images.length).keys()]);
    currentIndex = 0;
    showImage(shuffledIndexes[currentIndex]);
}

function shuffleArray(array) {
    for (let i = array.length - 1; i > 0; i--) {
        const j = Math.floor(Math.random() * (i + 1));
        [array[i], array[j]] = [array[j], array[i]];
    }
    return array;
}

// Initialize with the first image
showImage(shuffledIndexes[currentIndex]);
```

Folder Structure

Ensure your folder structure looks something like this:

```
project-folder/
|
├── index.html
├── script.js
└── pics/
    ├── image1.jpg
    ├── image2.jpg
    ├── image3.jpg
    └── image4.jpg
```

This should give you a functional random image generator! You can expand and modify the code according to your needs. Enjoy coding! 😀

References

1. github.com 2. github.com 3. github.com 4. github.com

You

Almost there, however, I want the images list to dynamically adapt to all image sin the pics folder e.g. if there are 10 images, all 10 image names should be in there, if 5 images are added before running again, these 5 images should now be included in the images list. Can you do that?

Copilot

Absolutely! To achieve this, you can use the Fetch API to dynamically load the image file names from the server. You would need a server-side script (e.g., using Node.js) to read the contents of the "pics" folder and return the list of images.

Here's a basic example using Node.js:

Node.js Server Script

First, create a new file (e.g., server.js) in your project folder with the following content:

```
const express = require('express');
const fs = require('fs');
const path = require('path');
const app = express();
const port = 3000;

// Serve static files from the project folder
app.use(express.static(__dirname));

app.get('/images', (req, res) => {
    const picsDir = path.join(__dirname, 'pics');
    fs.readdir(picsDir, (err, files) => {
        if (err) {
            return res.status(500).send('Unable to scan directory');
        }
        const imageFiles = files.filter(file => /\.(jpg|jpeg|png|gif)$/i.test(file));
        res.json(imageFiles);
    });
});

app.listen(port, () => {
    console.log(`Server is running at http://localhost:${port}`);
});
```

HTML and JavaScript

Update your index.html and script.js to fetch the images dynamically:

index.html

Make sure your script tag is updated to include the JavaScript code:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Random Image Generator</title>
    <style>
        #image-container img {
            width: 500px;
            height: auto;
        }
    </style>
</head>
<body>
    <div id="image-container">
        <img id="random-image" src="" alt="Random Image">
    </div>
    <button onclick="showPreviousImage()">Back</button>
    <button onclick="showRandomImage()">Next</button>
    <button onclick="randomizeOrder()">Randomize</button>
```

```
    <script src="script.js"></script>
</body>
</html>
```

script.js

Update your script to fetch the images from the server:

```
let images = [];
let currentIndex = 0;
let shuffledIndexes = [];

function fetchImages() {
    fetch('/images')
        .then(response => response.json())
        .then(data => {
            images = data.map(file => `pics/${file}`);
            shuffledIndexes = shuffleArray([...Array(images.length).keys()]);
            showImage(shuffledIndexes[currentIndex]);
        })
        .catch(error => console.error('Error fetching images:', error));
}

function showImage(index) {
    const imgElement = document.getElementById("random-image");
    imgElement.src = images[index];
}

function showPreviousImage() {
    currentIndex = (currentIndex - 1 + images.length) % images.length;
    showImage(shuffledIndexes[currentIndex]);
}

function showRandomImage() {
    currentIndex = (currentIndex + 1) % images.length;
    showImage(shuffledIndexes[currentIndex]);
}

function randomizeOrder() {
    shuffledIndexes = shuffleArray([...Array(images.length).keys()]);
    currentIndex = 0;
    showImage(shuffledIndexes[currentIndex]);
}

function shuffleArray(array) {
    for (let i = array.length - 1; i > 0; i--) {
        const j = Math.floor(Math.random() * (i + 1));
        [array[i], array[j]] = [array[j], array[i]];
    }
    return array;
}

// Initialize by fetching images from the server
fetchImages();
```

Folder Structure

Ensure your folder structure now looks something like this:

```
project-folder/
│
├── index.html
```

├── script.js

├── server.js

└── pics/

  ├── image1.jpg

  ├── image2.jpg

  ├── image3.jpg

  └── (other images...)

Running the Server

Make sure you have Node.js and npm installed. Then, in your project directory, run:
npm init -y
npm install express
node server.js
This should start your server on *http://localhost:3000* and dynamically fetch images from the "pics" folder.

Give it a try and let me know how it goes! 😀

References

1. github.com 2. github.com 3. github.com

You

Assume I can't use Node.js or any external server, I'm hosting the repo using GitHub pages

Copilot

Got it! Since we can't use server-side scripting, we'll have to use a different approach to dynamically manage the list of images. One way to achieve this is to manually maintain a JSON file that lists all the images. Every time you add new images to the "pics" folder, you'll need to update this JSON file.

Here's how you can do it:

Folder Structure

Ensure your folder structure looks something like this:

project-folder/

│

├── index.html

├── script.js

├── images.json

└── pics/

  ├── image1.jpg

  ├── image2.jpg

  ├── image3.jpg

  └── (other images...)

JSON File

Create a JSON file (e.g., images.json) that lists all the images in the "pics" folder:

```
[
    "pics/image1.jpg",
    "pics/image2.jpg",
    "pics/image3.jpg"
]
```

HTML and JavaScript

Update your index.html and script.js to load the images from the JSON file.

index.html

Ensure your script tag is updated to include the JavaScript code:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Random Image Generator</title>
    <style>
        #image-container img {
            width: 500px;
            height: auto;
        }
    </style>
</head>
<body>
    <div id="image-container">
        <img id="random-image" src="" alt="Random Image">
    </div>
    <button onclick="showPreviousImage()">Back</button>
    <button onclick="showRandomImage()">Next</button>
    <button onclick="randomizeOrder()">Randomize</button>

    <script src="script.js"></script>
</body>
</html>
```

script.js

Update your script to fetch the images from the JSON file:

```
let images = [];
let currentIndex = 0;
let shuffledIndexes = [];

function fetchImages() {
    fetch('images.json')
        .then(response => response.json())
        .then(data => {
            images = data;
            shuffledIndexes = shuffleArray([...Array(images.length).keys()]);
            showImage(shuffledIndexes[currentIndex]);
        })
        .catch(error => console.error('Error fetching images:', error));
}
```

```
function showImage(index) {
    const imgElement = document.getElementById("random-image");
    imgElement.src = images[index];
}

function showPreviousImage() {
    currentIndex = (currentIndex - 1 + images.length) % images.length;
    showImage(shuffledIndexes[currentIndex]);
}

function showRandomImage() {
    currentIndex = (currentIndex + 1) % images.length;
    showImage(shuffledIndexes[currentIndex]);
}

function randomizeOrder() {
    shuffledIndexes = shuffleArray([...Array(images.length).keys()]);
    currentIndex = 0;
    showImage(shuffledIndexes[currentIndex]);
}

function shuffleArray(array) {
    for (let i = array.length - 1; i > 0; i--) {
        const j = Math.floor(Math.random() * (i + 1));
        [array[i], array[j]] = [array[j], array[i]];
    }
    return array;
}

// Initialize by fetching images from the JSON file
fetchImages();
```

Updating Images

Whenever you add new images to the "pics" folder, make sure to update images.json with the new image paths. This way, your image list will dynamically adapt without needing server-side scripting.

Give this a try and let me know how it works for you! 😀

References

1. github.com 2. github.com 3. github.com

Conversation by Fifa Utuber with Copilot on Wednesday, Jan 15 2025